TRANSMISSION LINE FAULT DETECTION AND OVERCURRENT RELAY COORDINATION

Using MATLAB/Simulink

Project Report

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Abstract :

This project models transmission-line fault behavior and overcurrent relay coordination in MATLAB/Simulink. The system includes a three-phase source, two circuit breakers representing primary and backup protection, three-phase voltage/current measurement blocks, a transmission-line section, a load, a three-phase fault block, RMS measurement and relay logic. Five operating cases are considered: normal load, line-to-ground (L-G), line-to-line (L-L), double-line-to-ground (L-L-G), and three-phase (L-L-L) faults. The supplied project file includes current and voltage waveforms, IEC standard-inverse relay curves, a primary/backup coordination plot, trip-signal plots, and a results table. The results table records a normal-load current of approximately 105 A and fault-current values of approximately 700 A, 1,350 A, 1,400 A, and 1,580 A for the four fault cases. These are reported as the values in the supplied project file; they should be treated as simulation results for the configured model rather than field measurements.

1. Introduction

Power-system faults can cause excessive current, voltage depression, equipment stress, and loss of supply. Protective relays detect abnormal electrical quantities and issue trip commands to circuit breakers. The breaker isolates the faulted section so that the rest of the system can continue operating where possible. Overcurrent protection is widely used because the relay can respond when measured current exceeds a selected pickup value.

This project uses a primary relay and a backup relay. The primary relay is intended to clear a fault first. The backup relay is set with a longer operating delay so it can clear the fault if the primary protection or its breaker fails.

2. Problem Statement

The project addresses the detection of transmission-line faults and the coordination of primary and backup overcurrent protection in a MATLAB/Simulink model. The main task is to observe how fault type changes current and voltage waveforms, and to check whether relay logic produces the intended trip sequence.

3. Objectives

- Build a three-phase source, line, load, measurement, fault, and breaker model.
- Measure three-phase voltage and current during normal and faulted operation.
- Use RMS measurements to obtain phase-current magnitudes for overcurrent relay logic.
- Implement primary and backup IEC standard-inverse relay characteristics.

- Compare primary and backup relay operating behavior for five test cases.
- Generate waveform plots, relay-characteristic plots, coordination plots, and a results table.

4. Theory

4.1 Overcurrent pickup

An overcurrent relay operates when the measured current exceeds the pickup current. In the supplied relay implementation, the pickup setting is Is = 125 A. The phase RMS currents are compared, and the largest phase RMS current can be used as the operating quantity:

$$\mathbf{I_{max} = \max(I_{A,RMS}, I_{B,RMS}, I_{C,RMS})}$$

The current multiple is M = I/Is, where I is the current used by the relay and Is is the pickup current.

4.2 IEC standard-inverse characteristic

The IEC standard-inverse operating-time equation used in the supplied MATLAB code is:

$$\mathbf{t = 0.14 \times TMS / (M^{0.02} - 1)}$$

Here, t is the ideal relay operating time in seconds, TMS is the time-multiplier setting, and M is the current multiple. For current above pickup, the operating time decreases as the current multiple increases.

4.3 Primary and backup protection

The supplied MATLAB plotting code uses TMS_primary = 0.10 and TMS_backup = 0.25, with Is = 125 A. For the same current multiple, the higher backup TMS gives a longer ideal operating time. The coordination margin is calculated as t_backup − t_primary. This separation is intended to allow the primary relay and breaker to clear the fault before the backup protection operates.

4.4 Fault types

Fault case	Meaning	Fault block selection
Normal load	No fault is applied.	Fault block disabled
L-G	One phase connected to ground.	Phase A and Ground
L-L	Two phases shorted together.	Phases A and B; Ground disabled
L-L-G	Two phases connected together and to ground.	Phases A and B and Ground
L-L-L	All three phases shorted together.	Phases A, B and C; Ground disabled

5. System Model

The supplied Simulink diagram shows the following main power path: three-phase source → Breaker 2 (backup) → Three-Phase V-I Measurement (M2) → Breaker 1 (primary) → Three-Phase V-I Measurement (M1) → transmission-line section → three-phase load. A Three-Phase Fault block is connected at the protected-line section. A powergui block is included. Measurement signals feed the scopes and the relay logic.

Component	Role in the model
Three-phase source	Supplies the network.
Breaker 1 (B1)	Primary circuit breaker.
Breaker 2 (B2)	Backup circuit breaker.
Three-Phase V-I Measurement M1 and M2	Measures phase voltages and currents at two points.
Transmission-line block	Represents the protected line section.
Three-phase load	Represents downstream demand.
Three-Phase Fault	Applies the selected short-circuit condition.
RMS blocks and max block	Convert measured phase currents to RMS quantities and select the maximum.
Primary and backup relay functions	Apply pickup and time-delay logic and generate trip signals.
NOT blocks and scopes	Adapt trip logic to breaker control polarity and display signals.
powergui	Configures the Specialized Power Systems simulation environment.

5.1 Model parameters available in the project material

Parameter	Value / setting	Basis
Nominal line-to-line voltage	11 kV RMS	Project setup instructions supplied in the conversation
Frequency	50 Hz	Project setup instructions supplied in the conversation
Line length	20 km	Project setup instructions supplied in the conversation
Approximate load current	100 A	Initial setup target; actual normal-load result is reported as about 105 A
Relay pickup current, Is	125 A	MATLAB code in Project_2
Primary relay TMS	0.10	MATLAB code and primary characteristic plot
Backup relay TMS	0.25	MATLAB coordination plot
Fault switching times	0.1 s to 0.3 s	Fault-block screenshot supplied in the conversation
Fault resistance Ron	0.01 Ω	Fault-block screenshot supplied in the conversation

Ground resistance Rg	0.01 Ω	Fault-block screenshot supplied in the conversation
Snubber resistance Rs	1e6 Ω	Fault-block screenshot supplied in the conversation
Snubber capacitance Cs	inf	Fault-block screenshot supplied in the conversation
Solver / sample time	Exact solver step not recorded in the supplied project file	Confirm in the model Configuration Parameters before final submission

Note: The parameter values above combine the supplied Project_2 document with the project setup values shared during model development. Confirm the actual block dialogs and solver settings in the final .slx file before treating them as a complete parameter record.

6. Relay Implementation

The three measured phase currents are passed through RMS blocks. A max block selects the largest phase RMS current and sends it to the relay function. The relay receives the maximum current, a data input, and a reset input. A timer accumulates while current exceeds pickup; the relay asserts its trip output after the selected inverse-time operating condition is satisfied. A reset signal clears the timer and trip state.

The backup relay function shown in the supplied code uses Is = 125 A, TMS = 0.25, and Ts = 1e-4 s. The supplied MATLAB plot code uses TMS_primary = 0.10 and TMS_backup = 0.25 for the ideal IEC standard-inverse curves. The relay trip outputs are routed through NOT blocks to match the selected breaker control convention. Verify the polarity of each breaker control input in the actual Simulink library block.

The model also uses a primary-failure test arrangement: during backup tests, the primary breaker can be held closed deliberately so the backup relay is tested. For normal protection operation, restore the normal B1 trip connection after backup testing.

7. Test Methodology

The supplied project material calls for five separate runs. Before each run, return the model to its initial state, select the required fault type in the Three-Phase Fault block, run the simulation, and record the current, voltage, primary relay, backup relay, and breaker signals. The normal-load run is performed with the fault block disabled.

Test	Fault configuration	Switching time
Normal load	Fault disabled	None

L-G	Phase A to ground	0.1–0.3 s
L-L	Phases A and B; no ground	0.1–0.3 s
L-L-G	Phases A and B to ground	0.1–0.3 s
L-L-L	Phases A, B and C; no ground	0.1–0.3 s

For the primary-protection test, keep the normal B1 trip connection active. For the backup-protection test, temporarily prevent B1 from opening, while leaving B2 and the backup relay operational. Re-enable the original B1 connection after backup testing. Do not compare backup operating times from a run in which the backup relay was disabled or its breaker trip path was not connected.

8. Results and Discussion

The figures below are reproduced from the supplied Project_2 document. They show the waveform patterns and relay-characteristic plots included in that document. The result table is transcribed from its final page.

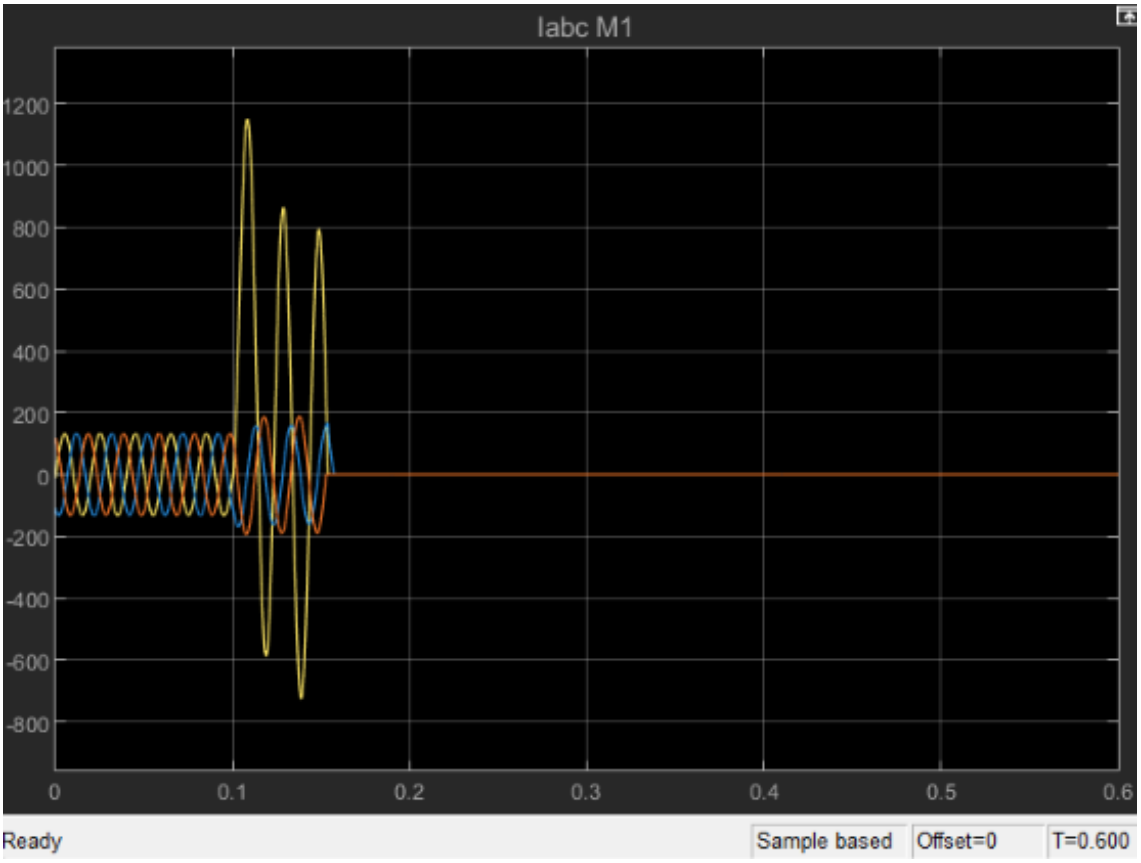


Figure 1. Three-phase current during a fault (M1).

The current plot shows the pre-fault phase currents, a transient rise when the fault is applied, and a reduction after the fault is cleared. The post-clearing current approaches zero at the measured point in this simulation.

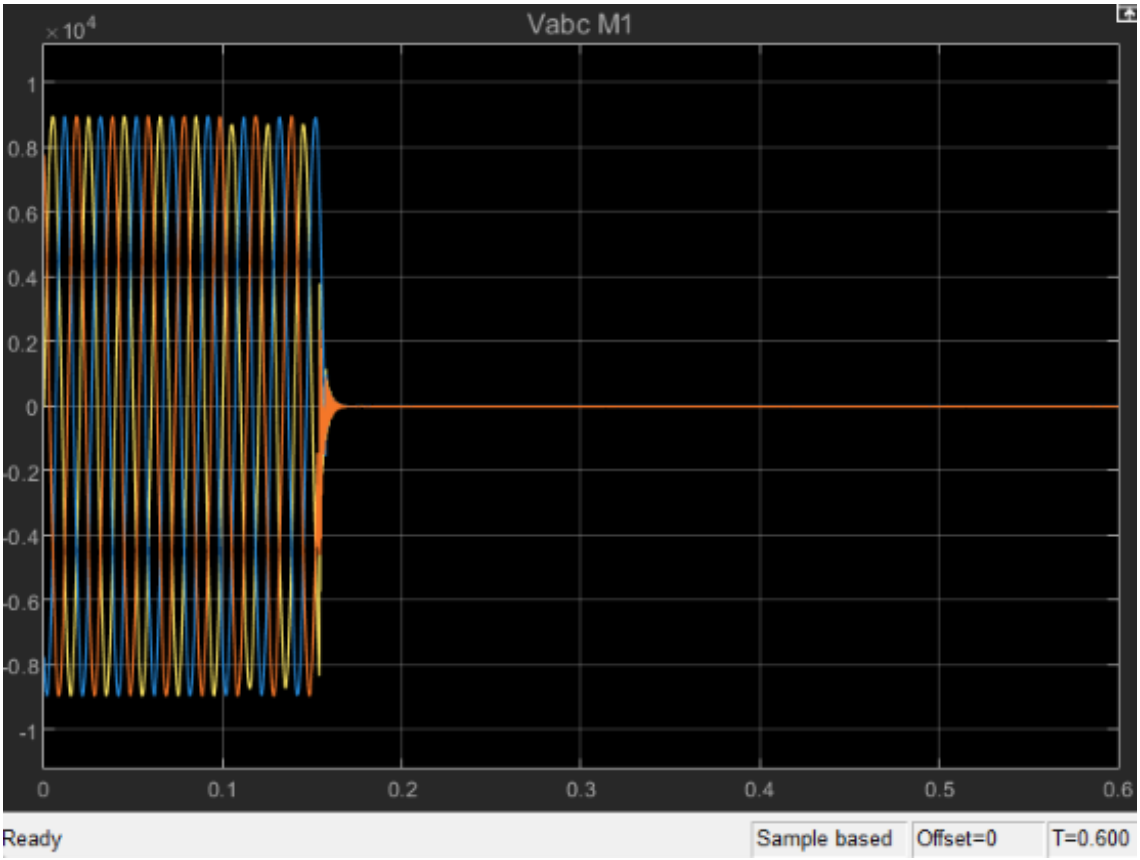


Figure 2. Three-phase voltage response (M1).

The voltage plot shows a disturbance around the fault and a reduction after isolation at the measurement point. The precise recovery behavior depends on the breaker action and the point at which voltage is measured.

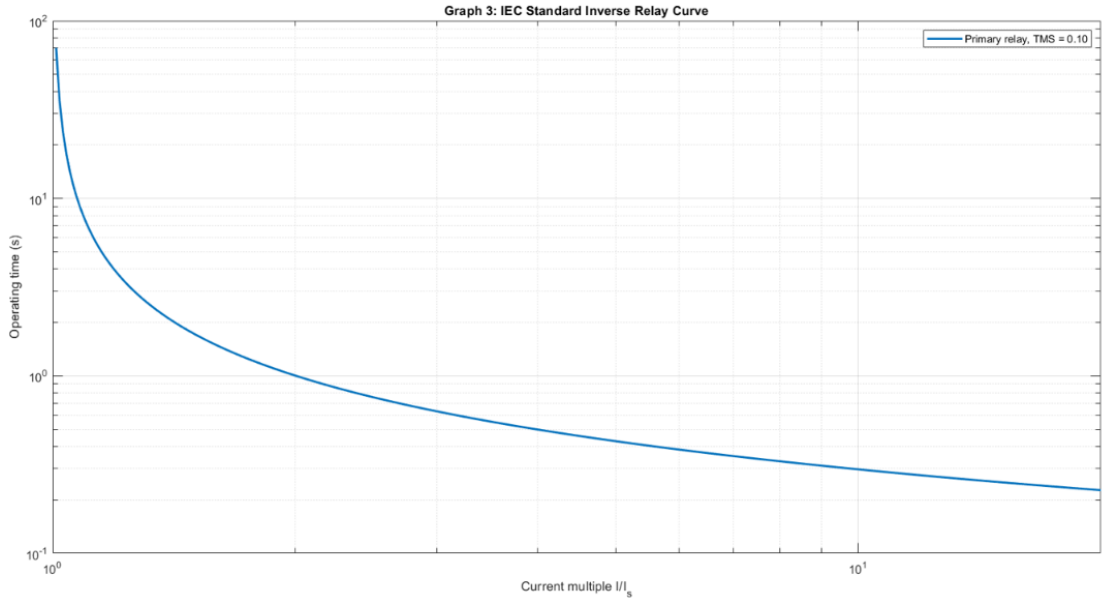


Figure 3. IEC standard-inverse relay operating-time curve.

The ideal IEC standard-inverse curve uses pickup current Is = 125 A and primary TMS = 0.10. The operating time decreases as the current multiple increases.

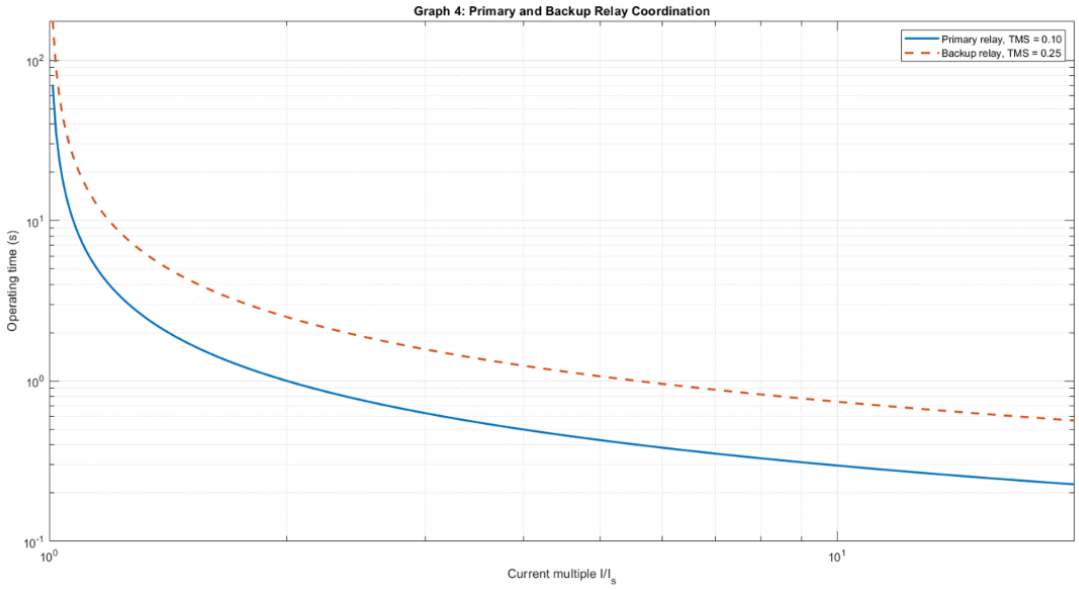


Figure 4. Primary and backup relay coordination.

The primary curve uses TMS = 0.10 and the backup curve uses TMS = 0.25. The backup curve lies above the primary curve, showing a longer ideal operating time for the same current multiple.

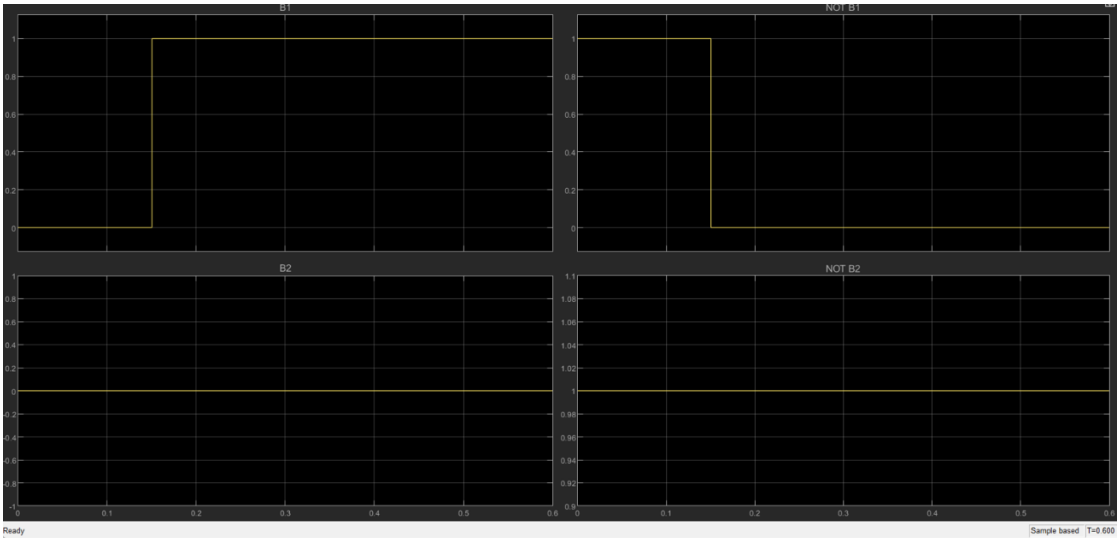


Figure 5. Trip signal and breaker response.

The trip-signal plot is intended to compare relay operation and breaker response. In interpreting it, distinguish the relay trip command from the breaker auxiliary/control signal and from the actual interruption of current.

8.1 Fault-case waveform figures

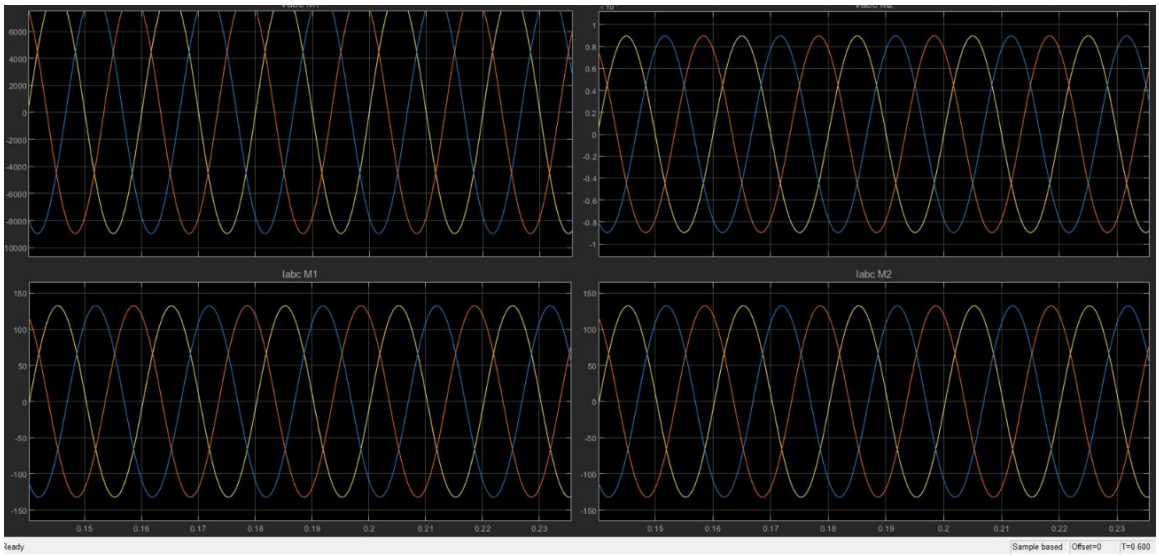


Figure 6. Normal-load voltage and current waveforms.

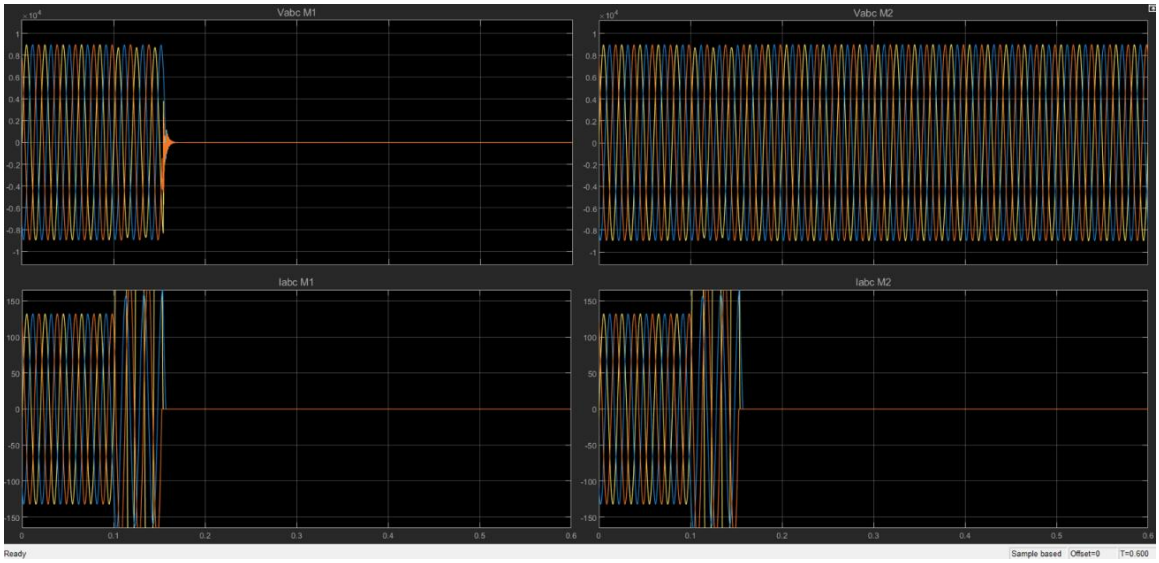


Figure 7. L-G fault waveforms.

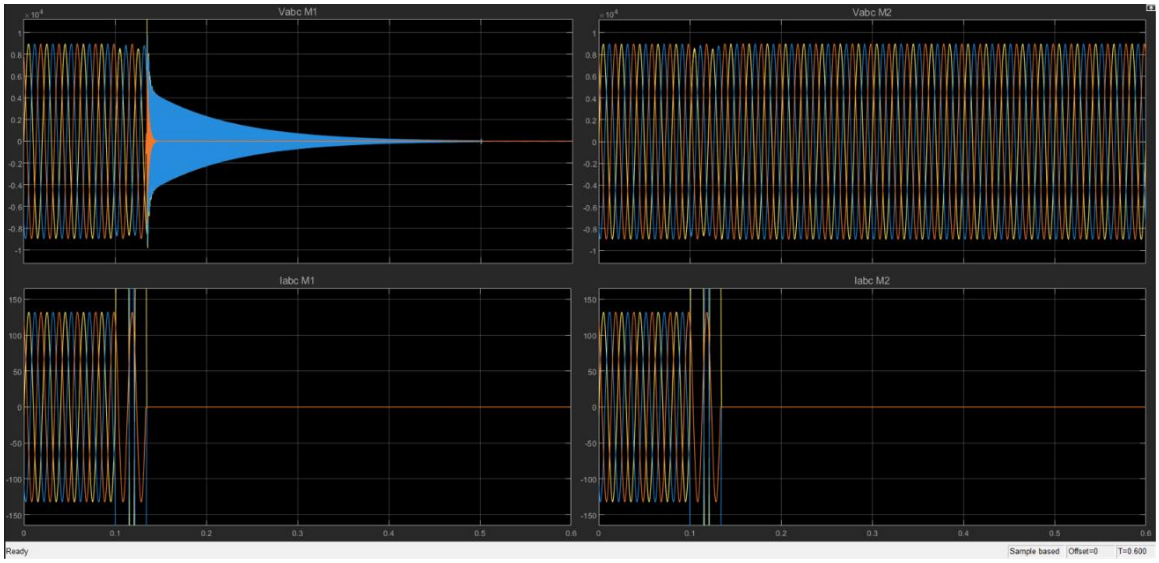


Figure 8. L-L fault waveforms.

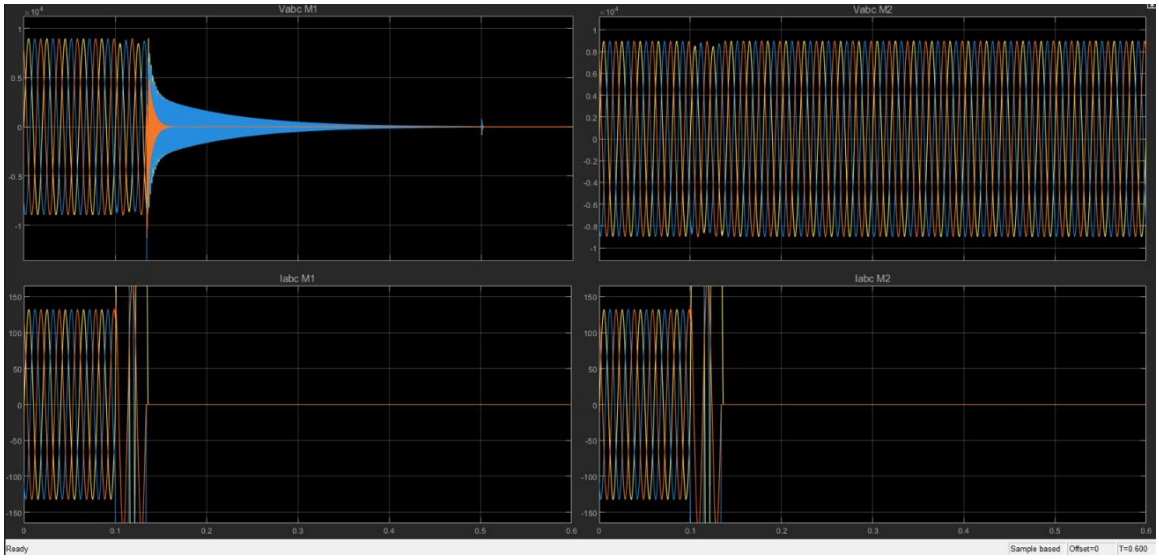


Figure 9. L-L-G fault waveforms.

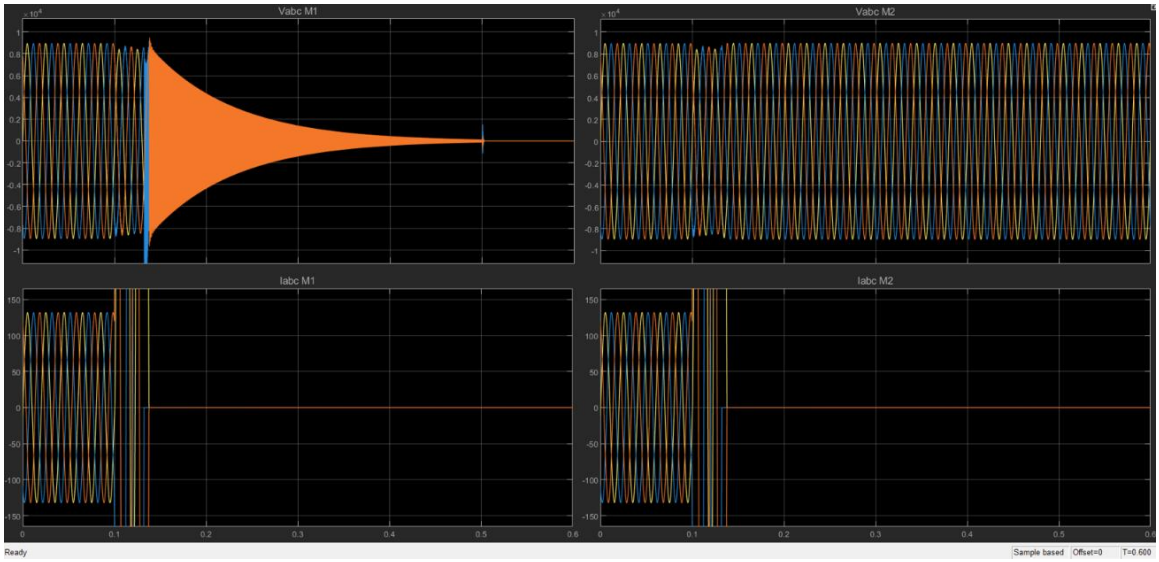


Figure 10. L-L-L fault waveforms.

8.2 Results table

The following values are reproduced from the results table on page 8 of the supplied Project_2 document. The current figures are shown as approximate model results. Relay times are the values recorded in that file.

Fault case	Fault current (A RMS)	Primary relay time (s)	Backup relay time (s)	Result
Normal load	≈ 105	No trip	No trip	Pass
L-G	≈ 700	0.050	1.18	Pass
L-L	≈ 1,350	0.035	0.85	Pass
L-L-G	≈ 1,400	0.030	0.80	Pass
L-L-L	≈ 1,580	0.030	0.80	Pass

Interpretation: The recorded fault-current magnitude increases from the L-G case through the L-L-L case in this configured model. The reported primary relay times are shorter for the higher-current fault cases, consistent with inverse-time overcurrent behavior. The backup relay times are longer than the primary times in the table, which is consistent with the intended coordination sequence.

Reporting note: The supplied document reports the values above, but the raw data files and exact timing cursors are not included. If the report is being assessed as a measured simulation study, verify the times from the saved simulation logs or Scope cursors and verify that each current value is an RMS value rather than a peak.

9. Validation

The supplied project includes analytical IEC standard-inverse curves calculated using t = 0.14 × TMS / (M^0.02 − 1), as well as simulation plots and a result table. This provides a basis for a qualitative comparison between inverse-time theory and the observed relay sequence. A complete numerical validation requires recording the actual current multiple, calculating the corresponding ideal relay time, and comparing it with the measured trip time from the simulation. The supplied file does not include the full set of current multiples or raw time series, so a numerical error calculation is not included here.

10. Limitations

- The model is a simulation and has not been validated against field measurements.
- The supplied results do not include raw logged data for independent recomputation of all RMS currents and trip times.
- The relay implementation uses simplified RMS measurement and timer logic; practical relays may use filtering and additional supervision.
- Current-transformer saturation, measurement errors, communication delays, and detailed breaker arc/interruption behavior are not documented in the supplied project.
- Exact solver configuration and sample time should be confirmed from the final Simulink model.

11. Future Scope

- Add distance protection for transmission-line backup.
- Add residual or earth-fault protection using zero-sequence current.
- Model current-transformer saturation and investigate its effect on relay operation.
- Include more detailed breaker dynamics and quantify the coordination time interval.
- Extend the model to communication-assisted protection or hardware-in-the-loop testing.
- Export logged RMS current, relay trip, breaker status, and voltage signals to MATLAB tables for reproducible results.

12. Conclusion

A MATLAB/Simulink model for transmission-line fault detection and overcurrent relay coordination was documented using the supplied Project_2 material. The model combines a three-phase source, line, load, fault insertion, measurement blocks, RMS-based overcurrent logic, primary and backup relays, and circuit breakers. The included plots illustrate fault-current transients, voltage disturbance, the IEC standard-inverse operating-time curve, primary/backup coordination, and trip-signal behavior. The supplied result table reports no trip during normal load and successful protection for the four fault cases. Final submission should include the saved Simulink model and, where possible, verified logged values for RMS currents and relay operating times.

References

1. MathWorks, *Simulink Documentation*. The MathWorks, Inc. Available at: https://www.mathworks.com/help/simulink/
2. MathWorks, *Overcurrent Relay Protection in AC Microgrid*. Available at: https://www.mathworks.com/help/simscape-electrical/ug/relay-overcurrent-protection.html
3. International Electrotechnical Commission (IEC), *IEC 60255-151:2009 — Measuring Relays and Protection Equipment: Functional Requirements for Over/Under Current Protection*. IEC, 2009.
4. MathWorks, *Simscape Electrical Documentation*. The MathWorks, Inc. Available at: https://www.mathworks.com/help/simscape-electrical/